



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009

PASSION, PURPOSE AND DRIVE.



BIOLOGY

Higher 1

Paper 2

8875/02

Thursday, 17 September

2 hours

Candidate Name: _____

Civics Group: _____ /08

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions

Section B

Answer any **one** question

At the end of the examination, fasten the answer papers securely and hand up separately.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
TOTAL	

This document consists of **13** printed pages.

Section A

For
Examiner's
Use

Answer **all** the questions in this section.

- 1** The electron micrograph in Fig. 1.1 shows an acinar cell found in the pancreas.

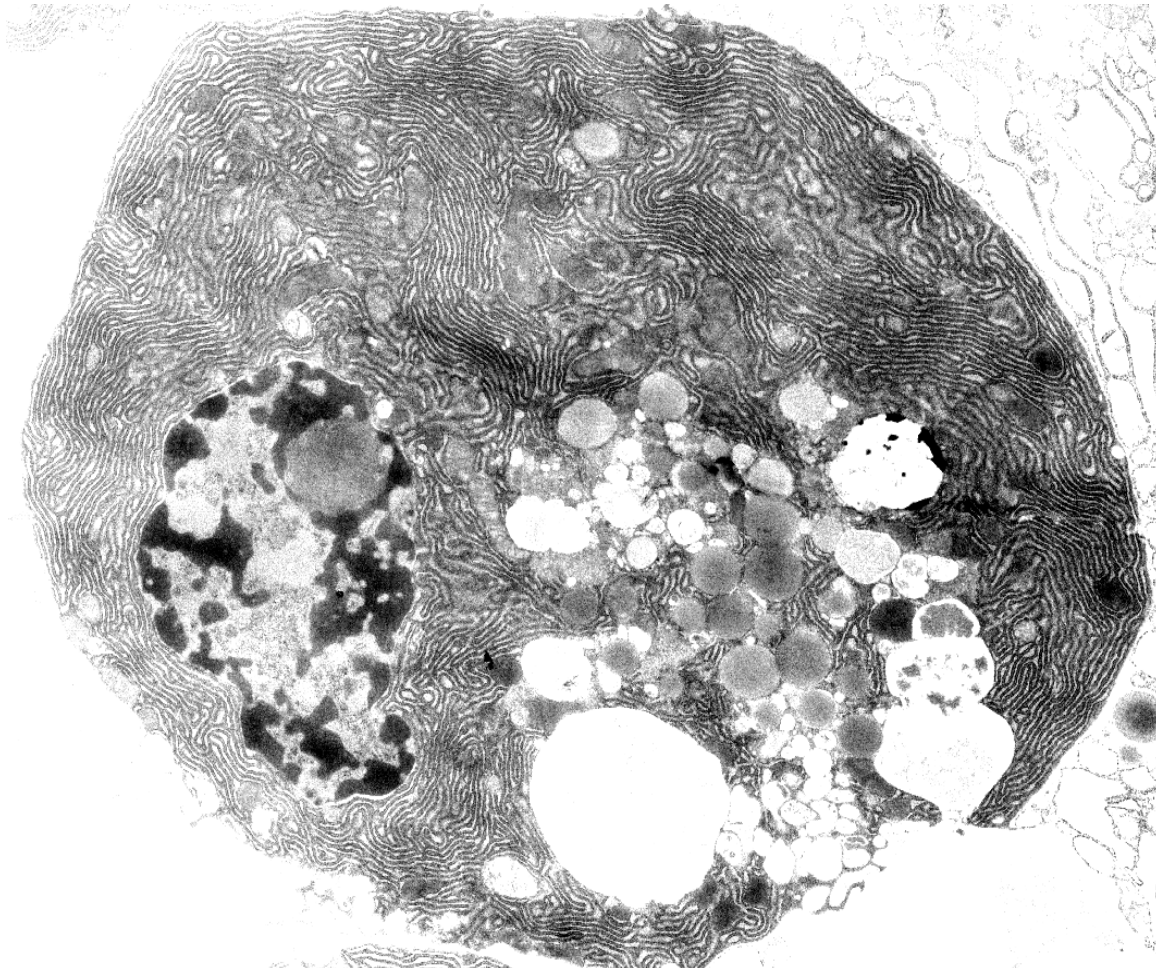


Fig. 1.1

- (a)** With reference to Fig. 1.1, suggest the principal function of the pancreatic acinar cell. Give **two** reasons to explain your answer.

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..... [3]

Fig. 1.2 shows two different structures, A and B, found in an animal cell.

For
Examiner's
Use

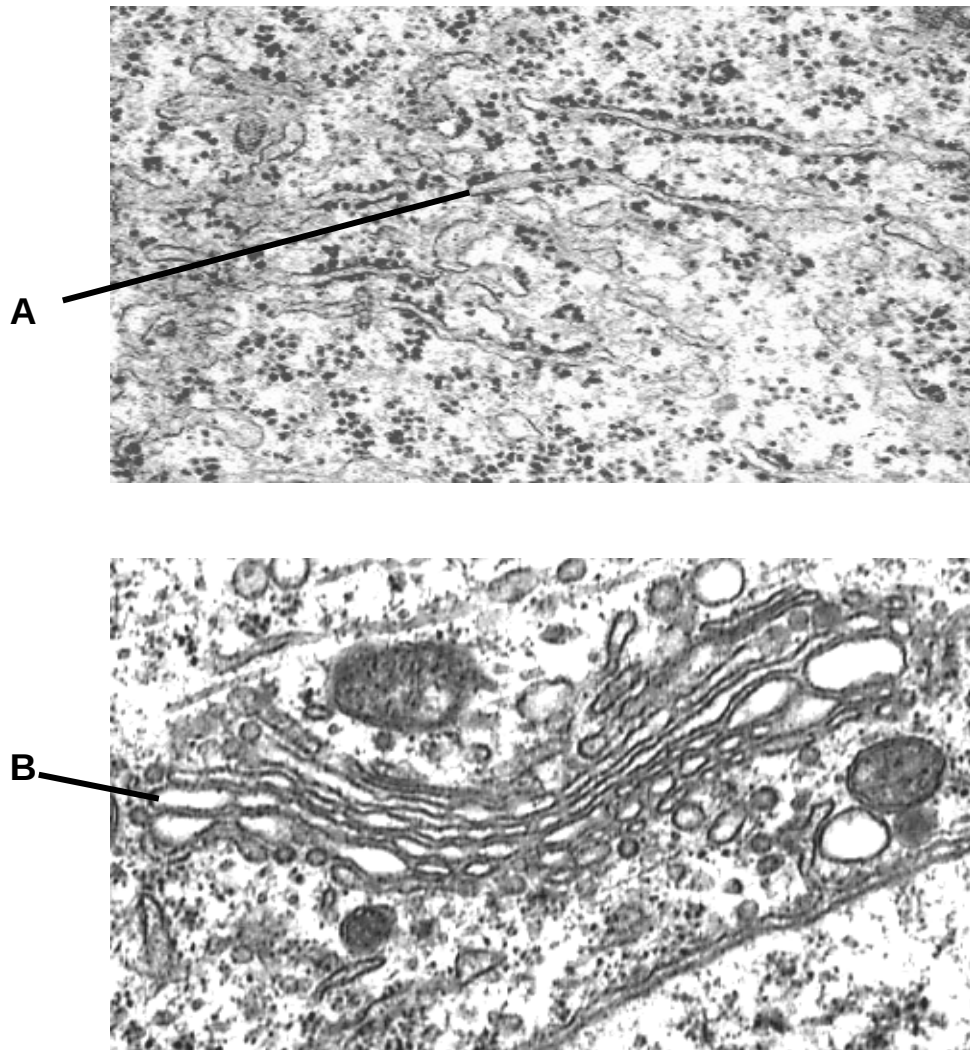


Fig. 1.2

(b) (i) Describe **two** ways in which A is similar to B.

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 [2]

(ii) The products of B are usually in the inactive form. Suggest the significance of this. [1]

.....
 [1]

- 2** Three different feather colours have been observed in Andalusian fowl – black, white and blue. Comparison of the alleles for white feathers and black feathers showed that there was a deletion of a single base-pair in the allele coding for white feathers.

To determine the mode of inheritance of feather colour in these fowl, several types of crosses were carried out.

In the parental generation, 2 crosses were carried out:

1. Female, pure-breeding black feathered fowl x male, pure-breeding white feathered fowl
2. Male, pure-breeding black feathered fowl x female, pure-breeding white feathered fowl

All of the F_1 generation in both crosses had blue feathers.

Sibling mating was then carried out within the F_1 generation, and the following results were obtained:

Black feathered fowl: 439

White feathered fowl: 442

Blue feathered fowl: 879

- (a)** State the type of cross carried out in the parental generation.

.....[1]

- (b)** Explain why a reciprocal cross was **not** carried out within the F_1 generation.

.....

[2]

- (c) Using suitable symbols, construct a genetic diagram to explain the results of the F_1 sibling mating.

*For
Examiner's
Use*

[5]

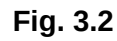
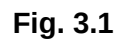
- (d) Construct a genetic diagram to predict the results of a cross between a blue feathered fowl and a white feathered fowl.

[4]

[Total: 12]

[Turn over

(a) With reference to Fig. 3.1 and Fig. 3.2, explain the role of light harvesting complex in both photosystems I and II.



.....[3]

- (b) Distinguish the electron transport chain in photophosphorylation from that in oxidative phosphorylation.

Photophosphorylation	Oxidative Phosphorylation
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[3]

The ATP and NADPH synthesized during photophosphorylation are utilised in the Calvin cycle for the synthesis of glucose, which is subsequently oxidised by the cell to produce ATP for its biochemical processes. Glucose may also be converted to sucrose and transported to the roots to be stored as starch.

- (c) Suggest a reason why ATP generated in the chloroplast is **not** used directly for the cell's biochemical processes.

.....
 [1]

[Turn over

(d) (i) In the space given, draw a monomer of starch.

[1]

(ii) Distinguish the molecule you have drawn in **(d)(i)** from a monomer of cellulose.

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 [1]

In animals, triglycerides serve as long term storage of energy, rather than carbohydrates.

(e) Suggest why triglycerides are better storage molecules than carbohydrates in animals.

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 [3]

[Total: 12]

- 4 Human growth hormone (hGH) is a peptide hormone containing 191 amino acids, synthesized and released by the cells of the anterior pituitary gland. The hormone stimulates growth and cell reproduction in humans. A deficiency of this hormone in children results in growth failure and short stature. In adults, growth hormone deficiency causes decreased muscle mass and poor bone density.

Fig. 4.1 shows how the gene for human growth hormone (hGH) can be transferred into a bacterium.

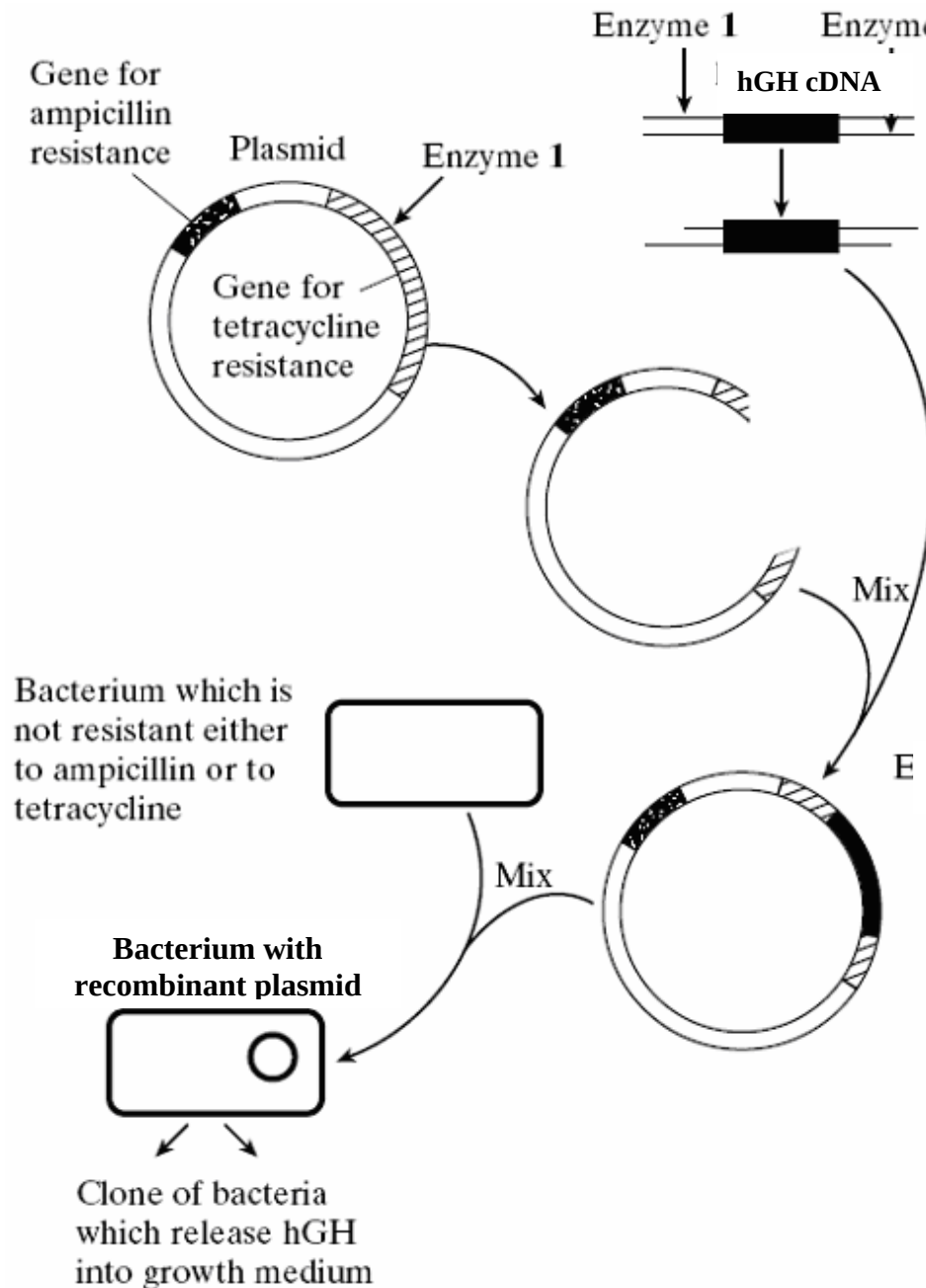


Fig. 4.1

(a) (i) Describe the process by which hGH cDNA is obtained.

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..... [3]

(ii) Explain why **Enzyme 1** is able to cut both the plasmid DNA and cDNA molecules.

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..... [2]

Phage viruses are commonly used as a means to transfer foreign DNA into bacteria cells. When a phage infects a layer or lawn of bacterial cells that are growing on a flat surface, a zone of lysis or growth inhibition can occur. This produces a clear zone in the bacterial lawn, known as a plaque. Each plaque originates from a single phage particle.

Fig. 4.2 illustrates the process involved in bacterial transformation and selection using phage vectors.

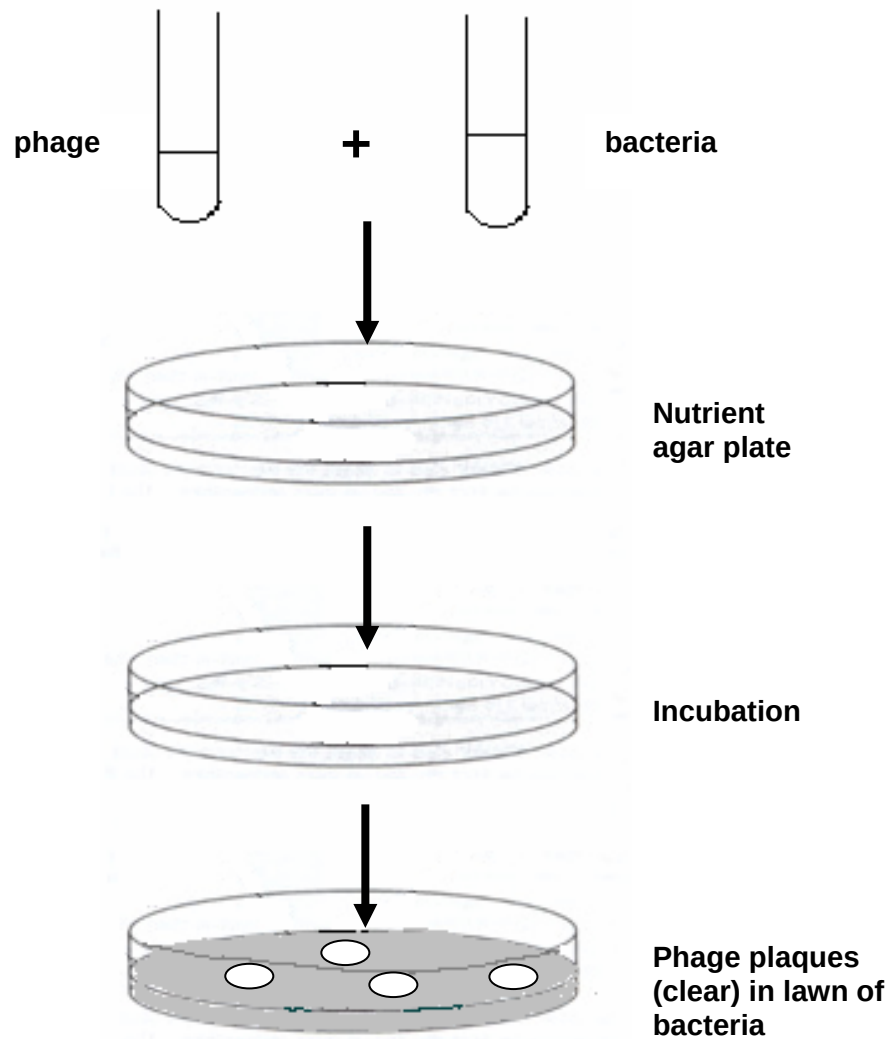


Fig 4.2

- (b) (i) With reference to a specific example, describe how a virulent phage can be used as a vector for gene cloning in bacteria.

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..... [4]

- (ii) Suggest **one** advantage of the use of phage vectors over plasmid vectors in transferring DNA into bacteria cells.

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..... [1]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5** **(a)** Describe the processing of a pre-mRNA in a eukaryote. [4]
- (b)** Explain how the molecular structure of HIV Protease is related to its functions. [8]
- (c)** Relate the features of adult stem cells to their use in the treatment of leukemia. [8]
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- 6** **(a)** Outline the differences between prokaryotic and eukaryotic control of gene expression. [8]
- (b)** Using a named example, explain how natural selection may bring about evolution. [6]
- (c)** Using specific examples, discuss the social and ethical issues of genetic engineering. [6]